# Prathibha Avvari

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 7.38 |
| Yadadri Junior College | Class XII (Telangana State Board of intermediate education) | 2020 | 97.70% |
| Jawahar Navodaya Vidyalaya | Class X (CBSE) | 2018 | 89.00% |

## THESIS

**Communication-Efficient Decentralized Federated Learning**

- **Cut communication volume 58%** by stacking **LinUCB layer/bucket bandits** in asynchronous decentralized FedML, reaching **77.2%** CIFAR-10 accuracy with **9.4 GB** vs.synchronous centralized FedAvg's **79.4%** with **22.5 GB.**
- **Cut training time 62% (16.6 h to 6.2 h)** vs. synchronous centralized FedAvg via **P2P gossip** in C++/LibTorch over raw **TCP**, with non-blocking connects and staleness-discounted aggregation so no client ever waits on stragglers.
- Validated as a **real-world distributed deployment** by running the same binary across **3 physical Linux servers** spanning three Ubuntu releases and compiler versions, scaling from **2 to 12 clients** and reaching **77.2%** accuracy.

## WORK EXPERIENCE

**IBM SkillsBuild and CSRBOX Academic Internship |** *Intern*

- **Trained** a Gradient Boosting Regressor on **10,682 real-world flight bookings**, reducing fare prediction error to **1,422 MAE** compared with a **3,676 baseline**, while achieving an **$R^2$ score of 0.72** with **robust performance**.
- **Developed** a multi-criteria travel ranker that evaluates **flight, train, bus, and car options** based on **cost, travel time, comfort, and CO emissions**, adapting recommendations to different traveller priorities and preferences.

## PROJECTS

**FedBench -- Communication-Aware FL Benchmarking** | *Github* | *LiveDemo*

- Exposed a **2× hidden communication cost** in SCAFFOLD (**157.9 vs 78.9 MB**, same **95% accuracy**) by implementing all **FedAvg**,**FedProx** and **SCAFFOLD** algorithms in PyTorch with per-round bandwidth tracking.
- **Cut a multi-hour manual experiment to one click** with an automated sweep retraining all three algorithms across **2-20 clients**, plotting accuracy, data, and time against both client count and communication round.
- **Built and deployed a public Streamlit dashboard** for reproducible in-browser experiments no install, no GPU..

**Kinesics Flash On System |** *Github* | *LiveDemo*

- Engineered a **CNN** gesture recognition model achieving **98% accuracy** across **10 classes** using **TensorFlow/Keras.**
- Developed **OpenCV**-based preprocessing to bridge infrared training data with live color webcam input.
- **Built and deployed** dual interfaces (**Tkinter desktop app and Streamlit web app**) with integrated Google Text-to-Speech, delivering an end-to-end real-time gesture-to-voice translation system, publicly deployed on Streamlit Cloud.

**Concrete Abridgement Vitality Prediction |** *Github*

- Cut error **5× ($R^2$ = 0.943, MAE 2.59 MPa)** by tuning **XGBoost** across **6 benchmarked models** with 5-fold CV.
- Got **88% of predictions within ±5 MPa** of lab values; SHAP identified age and water-binder ratio as key drivers.
- Built a **Keras** neural network (**$R^2$ = 0.938**) with input normalization, dropout and early stopping for regularization.

**Job Application Tracker** | *Github* | *LiveDemo*

- Built a fullstack job tracker with **Flask**, **SQLite**, **JS fetch API** for faster, error-free tracking over spreadsheets

- Architected **RESTful API endpoints** (GET/POST/PUT/DELETE) enabling real-time updates, improving usability
- **Deployed** the app on **Render** via **Gunicorn**, demonstrating solid production-ready deployment skills

## SKILLS

- **Languages:** C, C++, Python, Java, JavaScript, HTML/CSS
- **Libraries:** Pandas, NumPy, scikit-learn, XGBoost, Matplotlib, Seaborn, SHAP
- **Tools & Platforms:** Git, GitHub, Gunicorn, Render, Jupyter Notebook, Google Colab
- **Frameworks:** Flask, TensorFlow, Keras
- **Databases:** SQLite,mySQL

## RELEVANT COURSES

Data Structures and Algorithm ,Object-Oriented Programming ,Database Management System,Operating Systems,Computer Networks,Foundation of Machine Learning,Deep Learning

## POSITION OF RESPONSIBILITY

**Technical Club Coordinator** | JNTUHUCES

- Guided **100+** students; organized webinars, bootcamps, and a project expo alongside coding competitions.

**Technical Fest Coordinator** | JNTUHUCES

- Organized **8** technical events spanning coding, debugging, and paper and poster technical presentations sessions.

**NSS Coordinator** | JNTUHUCES

- Organized awareness rallies, competitions, and community cleaning drives, promoting civic engagement.

**Android Development , Git and GitHub Coordinator** | JNTUHUCES

- Led a hands-on Git and GitHub workshop for club members, improving their version-control and collaboration skills.

## EXTRACURRICULAR

- Content Creation, dancing,singing, cooking , playing chess, Taekwondo -Played upto Regional Levels